Understanding and Improving Compositionality of Stable Diffusion

Kavyanjali Agnihotri

Abstract

Modern text-to-image diffusion models struggle with compositional reasoning because standard text encoders lack strict spatial grounding. In this work, we investigate whether targeted text-encoder interventions can resolve these spatial deficits without the prohibitive computational cost of full-model retraining. We perform a modular "weight surgery" on Stable Diffusion 3.5 (SD3.5), replacing its default CLIP-L encoder with two compositionally-aware variants: CLIC RedCaps and CLIC CogVLM. Evaluating across a standardized benchmark of over 1,600 relational prompts using Qwen3-VL-30B and manual annotation, we demonstrate substantial improvements in spatial accuracy—particularly in complex directional and containment tasks—while improving global generative image fidelity (FID $\approx$ 57.9 compared to the baseline 60.6). Furthermore, our error analysis uncovers a "Reasoning-Output Paradox" within automated VLM evaluators and highlights the persistent dominance of semantic priors over structural nouns. Ultimately, our findings establish that visual compositionality relies heavily on text-encoder grounding rather than mere model scale, providing a highly efficient framework to enhance spatial reliability in production-ready diffusion pipelines.

1. Introduction

1.1. Problem Motivation

Modern text-to-image diffusion models have achieved remarkable success in generating high-quality, photorealistic images (Dhariwal & Nichol, 2021; Lugmayr et al., 2022; Poole et al., 2022; Saharia et al., 2022). However, they continue to struggle with compositional understanding (Chatterjee et al., 2024; Adamkiewicz et al., 2026), defined as the ability to correctly interpret and render spatial relationships, object interactions, and logical constraints specified in natural language prompts. While models such as Stable Diffusion 3.5 (SD3.5) (Esser et al., 2024; Rombach et al., 2022) can generate stunningly detailed images of individual objects, they systematically fail when prompts require precise spatial positioning, logical negation (Ban et al., 2024), or complex relational reasoning. This limitation represents a critical gap between language comprehension and visual generation, ultimately restricting the practical utility of these models for tasks that demand rigorous spatial control.

This research problem fundamentally stems from the nature of the training data used for standard Vision-Language Models (VLMs) (Bordes et al., 2024). Foundational models like CLIP (Radford et al., 2021) are frequently trained on vast datasets of loose, web-scraped image-caption pairs where strict spatial precision is rarely a priority. Historically, prior work has primarily focused on scaling diffusion architectures and optimizing general image quality metrics, such as Fréchet Inception Distance (FID) (Heusel et al., 2018), Importance Sampling (Grover et al., 2019), and standard CLIP scores (Hessel et al., 2022); while the systematic investigation of compositional failures has remained largely limited. Because most evaluation studies verify mere object presence rather than analyzing accurate spatial relationships, fundamental limitations in language-to-visual grounding are often masked. Consequently, the text embeddings fed into the diffusion transformer may inherently lack the explicit "spatial coordinates" necessary for the denoising process to arrange generated objects correctly.

Understanding and addressing compositional failures in diffusion models has several important implications:

1.1.1. Scientific Value

Compositionality is a foundational property of both human language and cognitive processing. Investigating precisely why state-of-the-art vision-language models fail at seemingly simple spatial reasoning tasks—such as accurately positioning a bed above a book—provides critical insights into the inherent limitations of current training paradigms and model architectures.

*Equal contribution . Correspondence to: KA <kavyanjali.agnihotri@student.uni-tuebingen.de>.

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1.1.2. Practical Value

Downstream applications requiring rigorous spatial precision (Ruiz et al., 2023; Tian et al., 2023; Hammoud et al., 2024; Voetman et al., 2023) are currently severely bottlenecked by the inability of underlying models to reliably interpret spatial language. Consequently, improving a model's compositional reasoning capabilities would significantly expand the reliability and utility of text-to-image systems, particularly within professional creative workflows and controlled generation pipelines.

1.2. Architectural Insight

By systematically isolating and replacing specific model components (namely, the text encoders) and comprehensively measuring the resulting impact, this work contributes to a deeper understanding of which architectural elements are primarily responsible for compositional understanding as opposed to general image quality. Crucially, our research focuses on determining whether specifically targeting the text encoding stage can resolve these spatial reasoning deficits without incurring the prohibitive computational costs of full model retraining.

1.3. Research Aim and Overview

In this work, we systematically quantify the failure rates across a diverse set of spatial and relational concepts within SD3.5. We specifically investigate whether replacing SD3.5's default CLIP-L encoder with CLIC (Compositional Language-Image Contrastive) (Peleg et al., 2025), an encoder explicitly trained on compositionally dense image-text pairs, measurably enhances the model's overall compositional reasoning capabilities.

To this end, we detail a comprehensive technical methodology for replacing the model's internal text encoder with two specialized variants: CLIC RedCaps[1], trained on 12 million Reddit image-text pairs rich in detailed spatial descriptions (Desai et al., 2021), and CLIC CogVLM[2], which is aligned with high-performance visual features (Wang et al., 2024). We analyze model performance across 13 distinct relational phrases (above, below, to the left of, to the right of, between, behind, closeby, far from, in front of, on top of, under, over, and without) by utilizing automated VLM-based evaluation via Qwen3-VL (Bai et al., 2023). Furthermore, we measure overall image quality using the Fréchet Inception Distance to verify that image fidelity is preserved despite the encoder swap. In addition to our empirical findings, we develop an automated, VLM-based evaluation infrastructure designed to assess these spatial relationships at scale, and we document the complete pipeline required for converting and integrating OpenCLIP models into production-level diffusion pipelines.

The remainder of this report is organized as follows: Section 2 details our data collection processes and architectural methodology; Section 3 presents both the quantitative and qualitative results of our experiments; and Section 4 discusses our core findings, acknowledges current limitations, and outlines directions for future work.

2. Methods and Data

The methodology for this research was structured as a multi-phase experimental pipeline designed to diagnose, address, and evaluate compositional reasoning deficits in state-of-the-art diffusion models. The following sections detail the iterative process of model selection, prompt engineering, architectural modification, and multi-modal evaluation.

2.1. Model Selection and Baseline Configuration

The study began with an extensive benchmarking phase to identify a robust foundation for compositional analysis. After evaluating several iterations of the Stable Diffusion family, Stable Diffusion 3.5 (SD3.5) Large Turbo[3] was selected due to its sophisticated triple-text-encoder architecture and high-fidelity "Turbo" distillation.

2.1.1. Deterministic Controls

To ensure experimental reproducibility and eliminate stochastic variance during comparative analysis, all image generations were locked to a fixed random seed of 42.

2.1.2. Hyperparameter Settings

Generations utilized a guidance scale (Zhan et al., 2026) of 4.0 and 28 inference steps, adhering strictly to the optimized defaults for the SD3.5 Turbo pipeline.

2.1.3. Output Format

Generated images were stored as raw PyTorch tensors (`float32`) and systematically normalized to ensure compatibility with downstream Vision-Language Model (VLM) assessment frameworks.

2.2. Exploratory Pilot and Problem Identification

Before formalizing the evaluation dataset, an unconstrained exploratory phase was conducted to map the specific failure

[1] https://huggingface.co/nmndeep/CLIC-ViT-L-14-224-PixPr-RedCaps

[2] https://huggingface.co/nmndeep/CLIC-ViT-L-14-224-CogVLM

[3] https://huggingface.co/stabilityai/stable-diffusion-3.5-large-turbo/blob/main/LICENSE.md

modes of the baseline SD3.5 model.

1. **Baseline Diagnostics:** Initial trials confirmed that while the model achieved exceptional photorealism, it suffered from obvious compositional failures. These were most notable in numerical reasoning (counting) (Li et al., 2025; Zhao et al., 2025) and in processing logical negations such as the phrase "not" (Ban et al., 2024).
2. **Spatial Blind Spots:** Systematic testing revealed persistent "blind spots" in basic vertical and horizontal stratification. For instance, prompts requiring objects to be "above" or "to the left of" one another frequently resulted in inverted coordinates or side-by-side placements.
3. **Relational Complexity:** We identified that SD3.5 struggled significantly with exclusion (e.g., "without") and physical interactions (e.g., "holding"), where objects were often inappropriately merged or rendered in complete isolation.

2.3. Dataset Construction: Self-Bench and Standardized Prompting

To explicitly isolate the linguistic understanding of the text encoder from the generative capacity of the diffusion transformer, we utilized the Self-Bench (Jeong et al., 2025) framework.

1. **Class Extraction:** We extracted 100 simple, visually distinct object classes (e.g., "giraffe", "book", "broccoli") to ensure that any failure in the final image could be confidently attributed to a misunderstanding of the spatial relationship, rather than an inability to render the objects themselves.
2. **Prompt Taxonomy:** A comprehensive benchmark of over 1,600 unique prompts was constructed across 13 core spatial and relational phrases: above, below, to the left of, to the right of, between, behind, closeby, far from, in front of, on top of, under, over, and without.
3. **Template Structure:** Every prompt followed a rigid syntactic template: "a photo of a [class1] [phrase] a [class2]". For the phrase "between", we adapted the template to: "a photo of a [class1] between [class2]s". This strict standardization allowed for controlled, apples-to-apples comparisons across the baseline and custom model variants.

2.4. Architectural Intervention: Encoder Replacement Pipeline

The central hypothesis of this work is that replacing the primary CLIP-L text encoder (Text Encoder 1) with a compositionally aware alternative can significantly improve the model's understanding of spatial semantics. We implemented a modular replacement of this component using CLIC (Compositional Language-Image Contrastive) variants, as depicted in Figure 1.

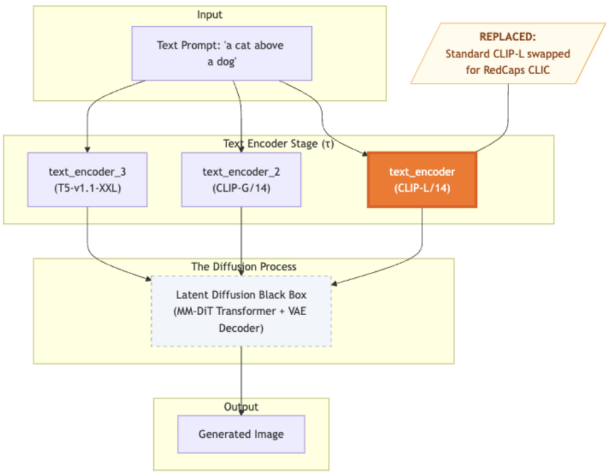


Figure 1. Architectural overview of the CLIP-L replacement with CLIC.

2.4.1. CLIC Variants: RedCaps and CogVLM

We integrated two distinct text encoders inherently optimized for compositional awareness:

1. **RedCaps CLIC:** Trained on 12 million image-text pairs sourced from Reddit, where alt-text frequently features highly descriptive spatial arrangements intended for accessibility.
2. **CogVLM CLIC:** A variant aligned with the CogVLM visual feature space, specifically designed to enhance the bridge between generative modeling and discriminative spatial understanding.

2.4.2. Technical "Weight Surgery"

Because the CLIC weights were distributed in the OpenCLIP format, they were architecturally incompatible out-of-the-box with the HuggingFace-based SD3.5 pipeline. To resolve this, we developed a custom conversion utility to perform the following transformations:

1. **QKV Matrix Decomposition:** The fused `in_proj_weight` matrix of shape 2304×768 was mathematically split into three separate Query (Q), Key (K), and Value (V) projections, each possessing a shape of 768×768.

2. **Activation Alignment:** The model configuration was explicitly updated to use the QuickGELU activation function, defined as $x \cdot \text{sigmoid}(1.702 \cdot x)$. We empirically observed that failure to align with QuickGELU led to immediate latent distribution collapse during generation.

3. **Dimension and Layer Mapping:** Weights were systematically mapped from OpenCLIP's `resblocks` nomenclature to HuggingFace's `encoder.layers` format, which included performing a necessary transpose operation on the final `text_projection` matrix.

2.4.3. Token Length

To satisfy standard Stable Diffusion architectural requirements, the `CLIPTokenizer` was strictly padded or truncated to a maximum token length of 77.

2.5. Multi-Prompt Evaluation

To analyze the resulting dataset of over 3,000 generated images, we initially evaluated both LLaVa (Liu et al., 2023) and Qwen3-VL. Our pilot analysis revealed that Qwen3-VL outputs aligned much more closely with manual human annotations compared to LLaVa (Chen et al., 2025) across a critical subset of phrases ("above", "below", "to the left of", "to the right of", and "between"). Consequently, we utilized the Qwen3-VL-30B-A3B-Instruct model[4] for the primary evaluation. To ensure robust assessment, we employed three distinct prompting strategies (Liu & Chilton, 2023) for the VLM evaluator:

1. **Direct Query:** A simple binary question regarding the spatial relationship, formatted as: *"Is the {class1} {phrase} the {class2}? Answer Yes, No, or Ambiguous."*

2. **Context-Prime (Expert Role):** Leveraging a system prompt with established context (Zhang et al., 2024): *"You are an Expert Visual Analyst and expert at spatial reasoning. 1. Identify the {class1} and describe its position. 2. Identify the {class2} and describe its position. 3. Based on their relative coordinates, is the {class1} {phrase} the {class2}? Conclusion: [Yes/No/Ambiguous]."* This multi-step mode successfully generated reasoning traces that provided deeper insights into VLM failure modes. For the phrase "between", the prompt was modified to: *"You are an expert at spatial reasoning. Task: Is the {class1} between the {class2}s? Step 1: Identify and locate the {class1}. Step 2: Identify the group of {class2}s or the boundary objects. Step 3: Analyze if the {class1} is positioned in the intermediate space separating the {class2}s. Conclusion: [Yes/No/Ambiguous]."*

3. **Inverse Logic:** Prompting the VLM to describe the relationship present in the image to cross-reference against the original generative prompt: *"You are an Expert Visual Analyst and expert at spatial reasoning. 1. Is the {class2} {inverse_phrase} the {class1}? 2. If so, then the {class1} is {phrase} the {class2}. Conclusion: [Yes/No/Ambiguous]."*

[4] https://huggingface.co/Qwen/Qwen3-VL-30B-A3B-Instruct

2.6. Manual Annotation and Fidelity Benchmarking

To ground the automated metrics and ensure VLM reliability, a comprehensive manual validation was conducted, comprising the following steps:

1. **Sampling Strategy:** For each phrase and model combination, a representative subset of 10 to 20 images was sampled for rigorous manual review.

2. **Labeling Scheme:** Manual annotation classified images into three distinct categories: Correct, Incorrect, or Ambiguous. This process allowed for the calculation of inter-evaluator agreement with the automated Qwen3-VL predictions.

3. **FID Calculation:** General image fidelity was measured using the Fréchet Inception Distance (FID) via the `pytorch-fid` library. We generated 5,000 images per model variant using standard COCO (Lin et al., 2015) validation captions. Following standard protocols, features were extracted using a pre-trained InceptionV3 network at the 2048-dimensional pooling layer with a batch size of 50. These distributions were then compared against the ground-truth COCO validation set to ensure that no overarching degradation in visual quality occurred as a side-effect of the encoder replacement.

2.7. Hardware and Computational Environment

All experiments were executed on the University of Tübingen's high-performance cluster, utilizing NVIDIA A100 (40GB) GPUs. The software environment was built utilizing `diffusers` version 0.24.0 and `transformers` version 4.35.0. The total computational footprint—including exploratory pilots, weight conversion, large-scale generation, and VLM inference—amounted to approximately 150 GPU hours.

3. Results

In this section, we present the quantitative and qualitative outcomes of our experiments. We compare the baseline

Stable Diffusion 3.5 Large Turbo (SD3.5) model against the CLIC RedCaps and CLIC CogVLM variants across a comprehensive set of spatial and relational benchmarks. Our analysis utilizes a multi-modal approach, combining large-scale automated VLM assessment, granular manual annotation to ensure evaluator reliability, and Fréchet Inception Distance for generative fidelity tracking.

3.1. Exploratory Pilot Findings

Prior to the structured benchmarking phase, our unconstrained pilot exploration revealed several notable failure modes inherent to the baseline SD3.5 model. Examples of these anomalies are illustrated in Figure 2.

1. **Semantic Prior Dominance:** The model frequently prioritized high-probability object tokens over structural or contextual nouns. For example, generating a literal pizza instead of interpreting the full concept of a "pizza flyer".
2. **Implicit Geometric Constraints:** Prompts involving complex structural objects, such as a "chessboard", consistently failed to adhere to the implicit 8×8 grid logic, resulting in irregular, non-Euclidean patterns.
3. **Action-Location Confusion:** In compositional prompts such as "A sketching X at Y", the model often rendered "A at X", "A at Y", or fundamentally confused the subject being sketched (X) with the physical location (Y) of the sketching action.
4. **Physical Interaction Failures:** Biological interactions, specifically intra-species or inter-species interactions, such as a "baby cow drinking milk", frequently resulted in fused anatomies or physically impossible object orientations.
5. **Optical Physics and Reflection:** The model struggled significantly with "half-reflection" tasks, demonstrating a lack of foundational understanding regarding the optical properties of mirrors and reflection.

3.2. Quantitative Analysis of Text Encoder Weights

Analysis of the surgical replacement revealed significant statistical shifts within the model's embedding space. The weights of the RedCaps and CogVLM text encoders demonstrated variances of 1.395×10^{-3} and 2.102×10^{-3} compared to the original CLIP-L encoder, respectively. This confirmed the successful integration and functional divergence of the replacement text encoders within our pipeline.

Furthermore, we performed a layer-by-layer weight divergence analysis between the original CLIP-L and the custom CLIC variants. Frobenius norms across the 12 transformer layers showed a consistently increasing trend from layer 0 to layer 11, with layer 11 exhibiting the largest total weight magnitude in both variants. Within the attention mechanisms, all projection matrices (Query, Key, Value, and Output) showed elevated norms in CogVLM relative to RedCaps, with Key projections ($1.21\times$ higher) and Value projections ($1.22\times$ higher) exhibiting the most pronounced differences. These elevations in attention projections suggest that the CogVLM variant may employ more fine-grained token relationships compared to the standard model. More detailed values across the layers of these weight differences is provided in Appendix 5.1.

3.3. Comparative Model Behavior and Manual Observations

Manual validation of 10 to 20 images per concept revealed distinctly different behavioral patterns between the standard baseline and the custom models.

3.3.1. Baseline SD3.5

From our investigation, we identified the following persistent limitations in the baseline model, as visualized in Figure 3:

1. **Positional Inversion and Order Bias:** The baseline performs poorly on the relation "above," but fails significantly more often on "below," as it frequently defaults to placing the first-mentioned object in the prompt above the second.
2. **Lack of Scene Diversity:** The model is largely unable to generate counter-intuitive scenes rarely found in natural training datasets, such as a "ball above book" or a "bench above cow".
3. **Directional Disparity:** While the model demonstrates adequate performance for "to the left of", it suffers a notable performance degradation when processing the inverse relation, "to the right of".
4. **Inverse Spatial Logic:** For spatial prepositions like "over" and "under", the model frequently generates objects exhibiting the exact inverse of the requested spatial relationship.

3.3.2. CLIC RedCaps and CogVLM

Compared to Standard SD3.5, we observed the following improvements and anomalies, with examples shown in Figure 4:

1. **Improved Verticality and Containment:** RedCaps significantly outperforms the baseline in vertical reasoning and containment tasks (e.g., "under", "over",



(a) "A pizza place flyer floating on water." (b) "A chessboard carved into stone." (c) "A person sketching a skyline on rooftop." (d) "A mirror reflecting only half of a person's face." (e) "A baby cow drinking milk from mother cow's udder."

Figure 2. Notable failure modes discovered during the unconstrained exploration phase.



(a) "A photo of a giraffe below a hair drier." (b) "A photo of a bench above a cow." (c) "A photo of a book above a sports ball." (d) "A photo of a toilet to the left of a potted plant." (e) "A photo of a toilet to the right of a potted plant."

Figure 3. Standard SD3.5 outputs demonstrating spatial failure modes.

"between"), although a minor fraction of images still exhibit generation errors.

2. **Reduction in Ambiguity:** Both custom variants produced markedly fewer ambiguous or entirely fused images compared to the standard model.

3. **CogVLM Performance Anomalies:** CogVLM behaves similarly to RedCaps overall but surprisingly performs worse on the phrases "to the right of" and "between" compared to RedCaps, despite still exceeding the baseline's performance. In certain edge cases, it replicates the exact spatial failures seen in Standard SD3.5.

Table 1. Manual annotation results across the different SD3.5 variants, formatted as (Ambiguous, Incorrect, Correct). The number of manually checked images is listed in parentheses next to the phrase.

Phrase	Standard	RedCaps	CogVLM
above (20)	(4,7,9)	(1,5,14)	(1,6,13)
below (20)	(4,8,8)	(1,12,7)	(0,9,11)
to the left of (20)	(4,5,11)	(0,9,11)	(0,8,12)
to the right of (20)	(1,17,2)	(0,9,11)	(0,11,9)
between (20)	(0,7,13)	(3,2,15)	(0,5,15)
behind (10)	(0,6,4)	(1,7,2)	(1,7,2)
closeby (10)	-	(0,1,9)	(0,1,9)
far from (10)	-	(0,7,3)	(0,6,4)
in front of (10)	(0,4,6)	(0,4,6)	(0,3,7)
without (10)	(0,7,3)	(0,6,4)	(1,5,4)

3.4. Quantitative Benchmark Results

Following the pilot, we utilized a standardized taxonomy derived from the Self-Bench classes to evaluate over 1,600 generated prompts. Final accuracy was determined through automated Qwen3-VL-30B scoring, cross-referenced against our manual annotations. The comprehensive results are summarized in Table 2.

3.5. Automated Evaluator Validation

3.5.1. LLaVA

Our initial automated evaluation across core spatial phrases (such as "above", "below", "to the left of", and "to the right of") revealed that LLaVA yielded significant inconsistencies, achieving an overall accuracy of just 56.4% (F1: 0.26). Specifically, on the phrase "above", LLaVA achieved a mere



(a) "A photo of a giraffe below a hair drier." (b) "A photo of a bench above a cow" (c) "A photo of a book above a sports ball." (d) "A photo of a toilet to the left of a potted plant." (e) "A photo of a toilet to the right of a potted plant."



(f) "A photo of a giraffe below a hair drier." (g) "A photo of a bench above a cow" (h) "A photo of a book above a sports ball." (i) "A photo of a toilet to the left of a potted plant." (j) "A photo of a toilet to the right of a potted plant."

Figure 4. RedCaps CLIC (top row) and CogVLM CLIC (bottom row) variant outputs.

Table 2. Comparative accuracy across SD3.5 model variants (measured in percentage).

Phrase	Standard	RedCaps	CogVLM
above	72	90	95
below	67	75	80
to the left of	80	85	85
to the right of	86	70	95
between	76	50	55
behind	65	60	60
closeby	-	90	80
far from	-	40	40
in front of	60	70	70
on top of	60	67	40
under	-	60	30
over	80	40	40
without	80	50	50

43.18% accuracy. Deeming this performance insufficient for reliable benchmarking, we pivoted to Qwen3-VL. This adjustment resulted in substantial improvements: our direct Qwen query approach achieved 73.48% overall accuracy (F1: 0.62), while the expert reasoning-based Qwen approach reached a comparable 71.46% accuracy (F1: 0.62).

3.5.2. QWEN3-VL-30B

We carefully measured the agreement between the Qwen3-VL-30B predictions and human evaluation. Table 3 presents a sample confusion matrix for the RedCaps SD variant (Phrase: "Above", $n = 20$), demonstrating high evaluator reliability. Table 4 illustrates the overall agreement between manual labels and Qwen3-VL accuracies across different relational phrases.

Table 3. Sample Confusion Matrix demonstrating the high alignment (+80% match) for the RedCaps model evaluating the "above" phrase utilizing the "expert reasoning" prompt strategy.

	QWEN Yes	QWEN No	QWEN Amb
Manual Yes	15	2	0
Manual No	0	3	0
Manual Amb	0	0	0

3.5.3. VLM LOGIC AND REASONING PATTERNS

Our deeper analysis of Qwen3-VL's internal reasoning traces identified critical failure bottlenecks in the evaluation pipeline. The VLM often reasons correctly about the visual elements but concludes with the incorrect logical answer. For example, regarding the image referenced in Figure 5, generated by the prompt "A photo of a bed under a book", Qwen3-VL correctly identifies both objects and describes their relative positions accurately:

"1. The bed is identified as the large surface covered with teal bedding, including sheets and pillows, located in the lower and middle portions of the image... 2. The book is an open book with visible text, resting on top of the bed... 3. The

Table 4. Qwen3-VL Agreement with Manual Annotations

Agreement Level	Standard SD3.5	RedCaps SD	CogVLM SD
High Match (+80%)	right of, over, without	above, left, closeby	above, below, right of
Decent (70-79%)	above, left, between	below, in front of	in front of
Poor Match (¡70%)	below, behind, on top	right of, between, under	between, behind, far from



Figure 5. An example of a Qwen3-VL failure mode where the model correctly identifies the objects and their relative physical positions but ultimately formulates an incorrect spatial conclusion.

book is physically located on top of the bed, not above it in a vertical sense. Therefore, the bed is not under the book in a spatial sense; rather, the book is on the bed. Conclusion: No."

Despite this accurate breakdown, it ultimately concludes that the bed is *not* under the book: a clear failure in spatial reasoning logic. This contradictory pattern was observed across multiple phrases. Furthermore, similar to the diffusion models themselves, the VLM evaluator frequently conflates "left" and "right". For the phrase "in front of", the VLM often rigidly expects direct physical occlusion, whereas our manual annotation accepted lateral foreground positioning.

Additionally, during inverse prompting modes, the VLM frequently produced excessively verbose reasoning outputs. When the final answer was cut off due to token limits, this resulted in an automatic "Ambiguous" label across all models, though this specific failure occurred most frequently when evaluating the standard SD outputs.

3.6. Image Fidelity (FID) Scores

To ensure that improving compositional alignment did not degrade overall image quality, FID scores were calculated against 5,000 COCO validation images. As shown in Table 5, the custom text encoders maintained parity with—and even marginally outperformed—the standard model's generative quality, likely due to more precise feature alignment during the denoising process.

Table 5. FID Scores Against COCO Ground-Truth Validation Set

Model	FID Score
Standard SD3.5	60.60
CLIC RedCaps	57.92
CogVLM Custom	57.86

4. Discussion & Conclusion

4.1. The Compositional Bottleneck in Diffusion Transformers

The gains in compositional accuracy, specifically an increase from 63 to 71 correct images for the phrase "to the left of" using RedCaps, and from 17 to 40 correct images for "to the right of" using CogVLM, provide a quantitative refutation of the prevailing assumption that generative models require full-model fine-tuning to improve structural adherence. Instead, our results suggest that the Diffusion Transformer (DiT) backbone in SD3.5 possesses a latent generative "competence" that remains underutilized due to impoverished conditioning signals from standard text encoders.

While standard CLIP-L is known to exhibit "bag-of-words" behavior that prioritizes mere object presence over relational syntax, the RedCaps CLIC encoder leverages Reddit's alt-text, which is frequently authored for accessibility purposes to provide granular spatial grounding for the visually impaired. This transition from noisy, web-crawled captions to descriptive, accessibility-focused text allows the text encoder to move beyond simple object-token mapping and toward a truly relational embedding space.

The CogVLM CLIC encoder shows a complementary, but distinct, behavioral profile. In our evaluation, CogVLM delivers especially strong gains on challenging directional composition (notably "to the right of"), indicating that encoder replacement can recover spatial distinctions that the baseline text encoder under-specifies. At the representation level, CogVLM exhibits consistently larger Frobenius norms than RedCaps across attention and MLP components, with the largest relative increase in `mlp.fc2.weight` and a clear elevation in Key, Value, and Output projections. Together, these trends suggest that CogVLM supplies a stronger, more targeted compositional conditioning signal

to the same SD3.5 DiT backbone, bypassing the need for computationally expensive backbone retraining.

4.2. The "Reasoning-Output Paradox" in VLMs

Our evaluation of Qwen3-VL-30B revealed a fundamental discrepancy between a model's internal reasoning traces and its final classification output, a phenomenon we term the *Reasoning-Output Paradox*.

4.2.1. Internal Representation vs. Classification Head

In multiple instances, Qwen3-VL generated a logically sound and physically accurate reasoning trace, yet concluded with an incorrect spatial answer. This suggests that the model's internal spatial awareness is significantly more advanced than its final classification head's ability to map that awareness to a rigid, categorical label.

4.2.2. Inverse Verbosity and Ambiguity

The tendency for Qwen3-VL to label standard SD3.5 outputs as "Ambiguous" more frequently in Inverse Prompting mode highlights a secondary failure mode: when the generated image itself lacks coherent spatial stratification, the VLM defaults to over-verbose descriptions that ultimately fail to trigger a definitive match with the target prompt's logic.

4.3. Deep Dive into Failure Modes: Semantic Priors vs. Structural Nouns

The exploratory phase of this project uncovered generation failures that point to a fundamental semantic hierarchy problem within the diffusion process itself.

4.3.1. The "Pizza Flyer" Phenomenon

When the model generates a literal pizza instead of a paper flyer, it visually illustrates that high-probability object tokens (e.g., "pizza") frequently override low-probability structural nouns (e.g., "flyer"). This suggests that the diffusion transformer's cross-attention layers remain heavily biased toward the most visually salient tokens in a prompt, regardless of their actual grammatical or modifying role.

4.3.2. Left-Right Asymmetry

The persistent confusion between "left" and "right", as well as "above" and "below", across all evaluated models, even after surgical encoder replacement, indicates a potential backbone-level bias. This demonstrates that even 70B+ parameter models struggle with directional absolute positioning, possibly due to the mirror-invariant properties of most underlying image-processing kernels and data augmentation strategies.

4.4. Surgical Insight: Weight Divergence in Layers

Our layer-wise analysis of the encoder replacement shows a broad increase in Frobenius norms from the early to late transformer blocks (layers 0–11), with the largest magnitudes concentrated in the final layers. This specific pattern indicates that the encoder substitution most meaningfully shifts the higher-level text representations utilized for DiT conditioning. Component-wise, CogVLM exhibits higher norms than RedCaps across both attention and MLP weights, with the largest relative increase located in `mlp.fc2.weight`. Within the attention mechanism itself, Key, Value, and Output projections are all elevated. These results are highly consistent with a modified, finer-grained token-interaction profile introduced after the encoder replacement.

4.5. Limitations and Future Outlook

Despite the demonstrable success of our modular intervention, certain constraints remain:

1. **Logical Negation:** Despite some measurable improvement with the concept of "without" when using the RedCaps and CogVLM SD3.5 variants, the broader problem of strict logical negation, previously documented in earlier Stable Diffusion models, remains largely unsolved.

2. **Vague Phrasing:** Relational phrases such as "far from", "closeby", and "on" proved to be highly subjective during evaluation, leading to the lowest levels of agreement between the automated VLM predictions and our manual human annotations.

3. **Hardware Efficiency:** While we optimized the evaluation pipeline for NVIDIA A100 clusters, the sheer computational cost of large-scale VLM assessment remains a significant overhead bottleneck for rapid, iterative testing in generative model development.

4.6. Conclusion

This research establishes that compositionality is not merely an emergent property of model scale, but rather a direct consequence of training data quality and text-encoder grounding. By replacing the standard CLIP-L encoder with compositionally-aware variants like RedCaps and CogVLM, we achieved substantial improvements in spatial reasoning while maintaining highly stable generative quality (FID 83.5). Ultimately, this "weight surgery" approach provides a robust, production-ready framework for enhancing the spatial reliability of text-to-image models without incurring the

prohibitive computational costs associated with full-model retraining.

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5. Appendix

5.1. Weight Comparison for RedCaps and CogVLM CLIC

Table 6 gives the Forbenius Norm of the difference between the original CLIP-L text encoder and the two CLIC variants layer by layer.

Table 6: Frobenius Norm of the difference between the original CLIP-L text encoder and the two CLIC variants layer by layer.

Layer Name	RedCaps CLIC	CogVLM CLIC
text_model.embeddings.position_embedding.weight	0.008173	0.010895
text_model.embeddings.token_embedding.weight	0.085015	0.108735
text_model.encoder.layers.0.layer_norm1.bias	0.00229	0.002786
text_model.encoder.layers.0.layer_norm1.weight	0.00792	0.008126
text_model.encoder.layers.0.layer_norm2.bias	0.003267	0.003584
text_model.encoder.layers.0.layer_norm2.weight	0.00894	0.00904
text_model.encoder.layers.0.mlp.fc1.bias	0.003521	0.004379
text_model.encoder.layers.0.mlp.fc1.weight	0.089096	0.111279
text_model.encoder.layers.0.mlp.fc2.bias	0.002044	0.0024
text_model.encoder.layers.0.mlp.fc2.weight	0.099352	0.121168
text_model.encoder.layers.0.self_attn.k_proj.bias	0.000566	0.000585
text_model.encoder.layers.0.self_attn.k_proj.weight	0.041081	0.049799
text_model.encoder.layers.0.self_attn.out_proj.bias	0.00209	0.002455
text_model.encoder.layers.0.self_attn.out_proj.weight	0.047979	0.059674
text_model.encoder.layers.0.self_attn.q_proj.bias	0.002145	0.002933
text_model.encoder.layers.0.self_attn.q_proj.weight	0.042259	0.04872
text_model.encoder.layers.0.self_attn.v_proj.bias	0.001954	0.002359
text_model.encoder.layers.0.self_attn.v_proj.weight	0.045594	0.056023
text_model.encoder.layers.1.layer_norm1.bias	0.002564	0.002799
text_model.encoder.layers.1.layer_norm1.weight	0.008256	0.00845
text_model.encoder.layers.1.layer_norm2.bias	0.003566	0.004009
text_model.encoder.layers.1.layer_norm2.weight	0.009192	0.009304
text_model.encoder.layers.1.mlp.fc1.bias	0.003676	0.004657
text_model.encoder.layers.1.mlp.fc1.weight	0.094367	0.118555
text_model.encoder.layers.1.mlp.fc2.bias	0.002028	0.002409
text_model.encoder.layers.1.mlp.fc2.weight	0.102252	0.126965
text_model.encoder.layers.1.self_attn.k_proj.bias	0.000493	0.000503
text_model.encoder.layers.1.self_attn.k_proj.weight	0.049575	0.063828
text_model.encoder.layers.1.self_attn.out_proj.bias	0.002076	0.00244
text_model.encoder.layers.1.self_attn.out_proj.weight	0.055471	0.069272
text_model.encoder.layers.1.self_attn.q_proj.bias	0.002261	0.002955
text_model.encoder.layers.1.self_attn.q_proj.weight	0.049195	0.062208
text_model.encoder.layers.1.self_attn.v_proj.bias	0.002267	0.002576
text_model.encoder.layers.1.self_attn.v_proj.weight	0.055425	0.069046
text_model.encoder.layers.10.layer_norm1.bias	0.002351	0.003011
text_model.encoder.layers.10.layer_norm1.weight	0.008342	0.008363
text_model.encoder.layers.10.layer_norm2.bias	0.002832	0.004221
text_model.encoder.layers.10.layer_norm2.weight	0.015948	0.016136
text_model.encoder.layers.10.mlp.fc1.bias	0.004668	0.006185
text_model.encoder.layers.10.mlp.fc1.weight	0.10911	0.12916
text_model.encoder.layers.10.mlp.fc2.bias	0.002667	0.003855
text_model.encoder.layers.10.mlp.fc2.weight	0.125197	0.159847
text_model.encoder.layers.10.self_attn.k_proj.bias	0.000441	0.000506
text_model.encoder.layers.10.self_attn.k_proj.weight	0.053107	0.062582
text_model.encoder.layers.10.self_attn.out_proj.bias	0.00243	0.003414
text_model.encoder.layers.10.self_attn.out_proj.weight	0.060872	0.06861
text_model.encoder.layers.10.self_attn.q_proj.bias	0.002373	0.003071
text_model.encoder.layers.10.self_attn.q_proj.weight	0.057276	0.069416

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Table 6: Frobenius Norm of the difference... (Continued)

Layer Name	RedCaps CLIC	CogVLM CLIC
text_model.encoder.layers.10.self_attn.v_proj.bias	0.002383	0.003037
text_model.encoder.layers.10.self_attn.v_proj.weight	0.062501	0.071082
text_model.encoder.layers.11.layer_norm1.bias	0.002951	0.003573
text_model.encoder.layers.11.layer_norm1.weight	0.008756	0.008711
text_model.encoder.layers.11.layer_norm2.bias	0.003578	0.005459
text_model.encoder.layers.11.layer_norm2.weight	0.008666	0.00925
text_model.encoder.layers.11.mlp.fc1.bias	0.006129	0.007344
text_model.encoder.layers.11.mlp.fc1.weight	0.129025	0.142604
text_model.encoder.layers.11.mlp.fc2.bias	0.003224	0.004483
text_model.encoder.layers.11.mlp.fc2.weight	0.144867	0.177204
text_model.encoder.layers.11.self_attn.k_proj.bias	0.000455	0.00053
text_model.encoder.layers.11.self_attn.k_proj.weight	0.053445	0.054786
text_model.encoder.layers.11.self_attn.out_proj.bias	0.003147	0.004476
text_model.encoder.layers.11.self_attn.out_proj.weight	0.058852	0.063449
text_model.encoder.layers.11.self_attn.q_proj.bias	0.002626	0.002632
text_model.encoder.layers.11.self_attn.q_proj.weight	0.057155	0.059257
text_model.encoder.layers.11.self_attn.v_proj.bias	0.002916	0.003503
text_model.encoder.layers.11.self_attn.v_proj.weight	0.056031	0.057719
text_model.encoder.layers.2.layer_norm1.bias	0.002412	0.00267
text_model.encoder.layers.2.layer_norm1.weight	0.008049	0.008201
text_model.encoder.layers.2.layer_norm2.bias	0.003513	0.00387
text_model.encoder.layers.2.layer_norm2.weight	0.012964	0.012956
text_model.encoder.layers.2.mlp.fc1.bias	0.003917	0.004983
text_model.encoder.layers.2.mlp.fc1.weight	0.098964	0.122465
text_model.encoder.layers.2.mlp.fc2.bias	0.001939	0.002318
text_model.encoder.layers.2.mlp.fc2.weight	0.101707	0.127526
text_model.encoder.layers.2.self_attn.k_proj.bias	0.000467	0.000496
text_model.encoder.layers.2.self_attn.k_proj.weight	0.051009	0.064013
text_model.encoder.layers.2.self_attn.out_proj.bias	0.001943	0.002303
text_model.encoder.layers.2.self_attn.out_proj.weight	0.058717	0.073337
text_model.encoder.layers.2.self_attn.q_proj.bias	0.002355	0.002809
text_model.encoder.layers.2.self_attn.q_proj.weight	0.051531	0.063047
text_model.encoder.layers.2.self_attn.v_proj.bias	0.002126	0.002457
text_model.encoder.layers.2.self_attn.v_proj.weight	0.05749	0.072235
text_model.encoder.layers.3.layer_norm1.bias	0.003124	0.003421
text_model.encoder.layers.3.layer_norm1.weight	0.008333	0.008608
text_model.encoder.layers.3.layer_norm2.bias	0.003646	0.003995
text_model.encoder.layers.3.layer_norm2.weight	0.014765	0.01487
text_model.encoder.layers.3.mlp.fc1.bias	0.004083	0.005219
text_model.encoder.layers.3.mlp.fc1.weight	0.103157	0.12407
text_model.encoder.layers.3.mlp.fc2.bias	0.001906	0.00239
text_model.encoder.layers.3.mlp.fc2.weight	0.107144	0.132704
text_model.encoder.layers.3.self_attn.k_proj.bias	0.000454	0.000484
text_model.encoder.layers.3.self_attn.k_proj.weight	0.057809	0.069354
text_model.encoder.layers.3.self_attn.out_proj.bias	0.001951	0.002359
text_model.encoder.layers.3.self_attn.out_proj.weight	0.062525	0.077634
text_model.encoder.layers.3.self_attn.q_proj.bias	0.002541	0.003468
text_model.encoder.layers.3.self_attn.q_proj.weight	0.058596	0.068339
text_model.encoder.layers.3.self_attn.v_proj.bias	0.002037	0.002453

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Table 6: Frobenius Norm of the difference... (Continued)

Layer Name	RedCaps CLIC	CogVLM CLIC
text_model.encoder.layers.3.self_attn.v_proj.weight	0.061898	0.076816
text_model.encoder.layers.4.layer_norm1.bias	0.002604	0.003014
text_model.encoder.layers.4.layer_norm1.weight	0.008217	0.008424
text_model.encoder.layers.4.layer_norm2.bias	0.003601	0.004041
text_model.encoder.layers.4.layer_norm2.weight	0.015312	0.015459
text_model.encoder.layers.4.mlp.fc1.bias	0.004018	0.005087
text_model.encoder.layers.4.mlp.fc1.weight	0.105542	0.126395
text_model.encoder.layers.4.mlp.fc2.bias	0.001791	0.002343
text_model.encoder.layers.4.mlp.fc2.weight	0.105839	0.133286
text_model.encoder.layers.4.self_attn.k_proj.bias	0.000382	0.000455
text_model.encoder.layers.4.self_attn.k_proj.weight	0.056201	0.069655
text_model.encoder.layers.4.self_attn.out_proj.bias	0.001882	0.002335
text_model.encoder.layers.4.self_attn.out_proj.weight	0.063063	0.080001
text_model.encoder.layers.4.self_attn.q_proj.bias	0.002408	0.003384
text_model.encoder.layers.4.self_attn.q_proj.weight	0.059726	0.073507
text_model.encoder.layers.4.self_attn.v_proj.bias	0.001827	0.002351
text_model.encoder.layers.4.self_attn.v_proj.weight	0.06099	0.076788
text_model.encoder.layers.5.layer_norm1.bias	0.002729	0.00334
text_model.encoder.layers.5.layer_norm1.weight	0.00811	0.008459
text_model.encoder.layers.5.layer_norm2.bias	0.003539	0.003892
text_model.encoder.layers.5.layer_norm2.weight	0.015581	0.015487
text_model.encoder.layers.5.mlp.fc1.bias	0.004108	0.005241
text_model.encoder.layers.5.mlp.fc1.weight	0.109652	0.130334
text_model.encoder.layers.5.mlp.fc2.bias	0.001734	0.002295
text_model.encoder.layers.5.mlp.fc2.weight	0.105196	0.133991
text_model.encoder.layers.5.self_attn.k_proj.bias	0.000397	0.000479
text_model.encoder.layers.5.self_attn.k_proj.weight	0.055773	0.068963
text_model.encoder.layers.5.self_attn.out_proj.bias	0.001737	0.00235
text_model.encoder.layers.5.self_attn.out_proj.weight	0.060264	0.080776
text_model.encoder.layers.5.self_attn.q_proj.bias	0.002164	0.003065
text_model.encoder.layers.5.self_attn.q_proj.weight	0.056652	0.069674
text_model.encoder.layers.5.self_attn.v_proj.bias	0.001693	0.00237
text_model.encoder.layers.5.self_attn.v_proj.weight	0.057974	0.077634
text_model.encoder.layers.6.layer_norm1.bias	0.002379	0.002759
text_model.encoder.layers.6.layer_norm1.weight	0.008284	0.008269
text_model.encoder.layers.6.layer_norm2.bias	0.00386	0.00423
text_model.encoder.layers.6.layer_norm2.weight	0.015301	0.015312
text_model.encoder.layers.6.mlp.fc1.bias	0.004155	0.005143
text_model.encoder.layers.6.mlp.fc1.weight	0.108709	0.130725
text_model.encoder.layers.6.mlp.fc2.bias	0.001783	0.002332
text_model.encoder.layers.6.mlp.fc2.weight	0.105817	0.135144
text_model.encoder.layers.6.self_attn.k_proj.bias	0.00046	0.000528
text_model.encoder.layers.6.self_attn.k_proj.weight	0.056081	0.068587
text_model.encoder.layers.6.self_attn.out_proj.bias	0.001763	0.002342
text_model.encoder.layers.6.self_attn.out_proj.weight	0.061265	0.078309
text_model.encoder.layers.6.self_attn.q_proj.bias	0.002066	0.002653
text_model.encoder.layers.6.self_attn.q_proj.weight	0.058913	0.069418
text_model.encoder.layers.6.self_attn.v_proj.bias	0.001692	0.002261
text_model.encoder.layers.6.self_attn.v_proj.weight	0.059581	0.076724

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Table 6: Frobenius Norm of the difference... (Continued)

Layer Name	RedCaps CLIC	CogVLM CLIC
text_model.encoder.layers.7.layer_norm1.bias	0.002778	0.00314
text_model.encoder.layers.7.layer_norm1.weight	0.008431	0.00839
text_model.encoder.layers.7.layer_norm2.bias	0.00349	0.003684
text_model.encoder.layers.7.layer_norm2.weight	0.015476	0.01543
text_model.encoder.layers.7.mlp.fc1.bias	0.004086	0.005151
text_model.encoder.layers.7.mlp.fc1.weight	0.10613	0.128304
text_model.encoder.layers.7.mlp.fc2.bias	0.001893	0.002423
text_model.encoder.layers.7.mlp.fc2.weight	0.10737	0.134731
text_model.encoder.layers.7.self_attn.k_proj.bias	0.000425	0.000508
text_model.encoder.layers.7.self_attn.k_proj.weight	0.053459	0.064851
text_model.encoder.layers.7.self_attn.out_proj.bias	0.001848	0.002337
text_model.encoder.layers.7.self_attn.out_proj.weight	0.060451	0.076152
text_model.encoder.layers.7.self_attn.q_proj.bias	0.002237	0.002805
text_model.encoder.layers.7.self_attn.q_proj.weight	0.056506	0.068229
text_model.encoder.layers.7.self_attn.v_proj.bias	0.001787	0.002218
text_model.encoder.layers.7.self_attn.v_proj.weight	0.058207	0.075241
text_model.encoder.layers.8.layer_norm1.bias	0.002233	0.002667
text_model.encoder.layers.8.layer_norm1.weight	0.008338	0.008513
text_model.encoder.layers.8.layer_norm2.bias	0.003256	0.003737
text_model.encoder.layers.8.layer_norm2.weight	0.015716	0.015836
text_model.encoder.layers.8.mlp.fc1.bias	0.004069	0.005309
text_model.encoder.layers.8.mlp.fc1.weight	0.106005	0.127332
text_model.encoder.layers.8.mlp.fc2.bias	0.00206	0.002734
text_model.encoder.layers.8.mlp.fc2.weight	0.112115	0.141943
text_model.encoder.layers.8.self_attn.k_proj.bias	0.000412	0.000481
text_model.encoder.layers.8.self_attn.k_proj.weight	0.051519	0.067374
text_model.encoder.layers.8.self_attn.out_proj.bias	0.002003	0.002552
text_model.encoder.layers.8.self_attn.out_proj.weight	0.059103	0.071199
text_model.encoder.layers.8.self_attn.q_proj.bias	0.002087	0.002898
text_model.encoder.layers.8.self_attn.q_proj.weight	0.053	0.069356
text_model.encoder.layers.8.self_attn.v_proj.bias	0.001918	0.002482
text_model.encoder.layers.8.self_attn.v_proj.weight	0.058047	0.071918
text_model.encoder.layers.9.layer_norm1.bias	0.003024	0.003472
text_model.encoder.layers.9.layer_norm1.weight	0.007936	0.008213
text_model.encoder.layers.9.layer_norm2.bias	0.003413	0.003914
text_model.encoder.layers.9.layer_norm2.weight	0.015666	0.015855
text_model.encoder.layers.9.mlp.fc1.bias	0.004386	0.005424
text_model.encoder.layers.9.mlp.fc1.weight	0.108228	0.126033
text_model.encoder.layers.9.mlp.fc2.bias	0.002245	0.003108
text_model.encoder.layers.9.mlp.fc2.weight	0.116033	0.146626
text_model.encoder.layers.9.self_attn.k_proj.bias	0.000478	0.000577
text_model.encoder.layers.9.self_attn.k_proj.weight	0.05651	0.062531
text_model.encoder.layers.9.self_attn.out_proj.bias	0.002153	0.002809
text_model.encoder.layers.9.self_attn.out_proj.weight	0.064986	0.072812
text_model.encoder.layers.9.self_attn.q_proj.bias	0.002454	0.002814
text_model.encoder.layers.9.self_attn.q_proj.weight	0.058875	0.066071
text_model.encoder.layers.9.self_attn.v_proj.bias	0.002173	0.002791
text_model.encoder.layers.9.self_attn.v_proj.weight	0.065208	0.074562
text_model.final_layer_norm.bias	0.003565	0.005115

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Table 6: Frobenius Norm of the difference... (Continued)

Layer Name	RedCaps CLIC	CogVLM CLIC
text_model.final_layer_norm.weight	0.00607	0.006321
text_projection.weight	29.918453	29.918283